Junghun Hwang (황정헌)

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Education

Konkuk University

Mar 2022 - Expected Feb 2028
B.S. in Smart Vehicle Engineering | Seoul, Korea
In progress: AI Mobility Systems

Research Interests

Autonomous-driving perception and decision-making, with an emphasis on learning-based and end-to-end policies and their closed-loop behavior under occlusion and input delay.

Research & Project Experience

Kookmin Autonomous Driving Competition - AI-Edge Challenge

Apr 2026 - Aug 2026
Team Lead | Team of 5

- Built and integrated a ROS 2 steering pipeline spanning data collection, imitation-learning training, TorchScript inference, and switching between learned and rule-based control.
- Prepared and reviewed Roboflow datasets and trained a unified YOLO detector for traffic lights, vehicles, obstacles, and cones.
- Integrated mission-management and driving modules and tuned the competition vehicle. The team used rule-based control in the final competition after persistent oscillation in the learned steering policy.
- Defined parking waypoints on a SLAM map and implemented reverse recovery logic for the parking event.

Team results: **Preliminary 2nd Place** | Finals 13th Place (course completed) | **Parking Special Award**

2026 Student AI · SW Mobility Competition

Mar 2026 - Ongoing
Team of 27 | Decision-Making Subteam (5)

- Developed ROS 2 path-planning and AEB logic; proposed and implemented Pure Pursuit–Stanley tracking, and authored the high-level control section of the team technical report.
- In the initial matched Gazebo comparison at 6 m/s, hybrid tracking reduced heading-error p95 by approximately **14%** and steering-command total variation by **10%** versus Stanley alone.
- Diagnosed a coordinate–timestamp inconsistency in cached-path reuse, implemented odometry-based reprojection, and added regression tests; the fix was incorporated into the team's main branch.
- Analyzed four recorded drives to compare planning horizon, return distance, curvature, and obstacle clearance using synchronized MCAP/Foxglove visualizations.
- Operated the vehicle during field tests of the updated software, confirming successful obstacle avoidance and return to the reference route.
- Completed real-vehicle waypoint runs on both preliminary and final courses at **10 m/s (36 km/h)** during venue practice.

Camera-Based V2I Cooperative End-to-End Autonomous Driving

2026 - Ongoing
Capstone Design Project | Team of 5 | Research & Design Phase

- Reviewed UniV2X's vehicle–infrastructure fusion framework and open-loop evaluation limitations, and prepared a literature summary to inform project scoping.
- Proposed a simplified design combining infrastructure object information with camera-based end-to-end planning. Contributed to scoping a subsequent closed-loop comparison on a small-scale vehicle; current work is limited to research and design.

UAV Waypoint Mission - ROS 2 / PX4 SITL

May 2026 - Jun 2026
Autonomous System Platform Course | UAV Waypoint Development | Team of 4

- Developed a ROS 2 mission package with CSV waypoints, TF-based arrival checks, and coordinate-mode handling; integrated it with an existing PX4 offboard controller via PoseStamped commands.
- Demonstrated sequential waypoint tracking and final autonomous landing in the PX4 SITL/Gazebo course simulation.

ROS 2 / CARLA Decision and Control

Mar 2026 - Jun 2026
Dream Semester Project | Decision-Making | Team of 3

- Implemented ROS 2 decision-making, Pure Pursuit waypoint tracking, and longitudinal control with CARLA integration and later vehicle-interface adaptation.
- Investigated simulated tracking instability by adjusting steering direction, lookahead, smoothing, and command limits; instability remained unresolved, with no real-vehicle run.

Awards & Competitions

2026 Kookmin AI-Edge Challenge
Parking Special Award (team)

2026 Hyundai Mobis Technical Special Award
Autonomous Driving Division | Team K.A.I. (team award)
Feb 10, 2026 | Oral presentation, jointly credited to Woojin Hwang and Junghun Hwang: *Decision Logic: Combining Pure Pursuit and Stanley for Scenario-Adaptive Control*.

2022 Drone Clash, General Division
3rd Place

Technical Skills

Languages: Python, C++

Autonomous Systems: ROS 2, CARLA, Gazebo, PX4 SITL

AI / Computer Vision: PyTorch, YOLO, OpenCV, Roboflow

Tools / Environment: Ubuntu, Git/GitHub, Foxglove, RViz2, rosbag2, QGroundControl